Fiscal Year:	FY 2021	**Task Last Updated:**	FY 11/04/2021
PI Name:	Avedisian, C. Thomas Ph.D.		
Project Title:	Use of FLEX Data to Enhance Video Image Processing for Studying Combustion of Soot-Producing Jet Fuel Droplets and Their Blends with Bio-Derived Fuels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015-16 Physical Sciences NNH15ZTT001N-15PSI-C: Use of the NASA Physical Sciences Informatics System – Appendix C
Start Date:	01/15/2018	**End Date:**	01/14/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/14/2021 per NSSC information (Ed., 10/22/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Reeves, Anthony Ph.D. (Cornell University)		
Grant/Contract No.:	80NSSC18K0480		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One of the most important challenges of our time is to mitigate the harmful impacts of particulate emissions (soot) from combustion engines on global climate change, and to reduce the consumption of liquid petroleum fuels that power transportation systems. A promising concept is to mix petroleum fuels with bio-derived fuels. An evaluation of the performance benefits rests on diagnostics to unravel the complex relationships between soot formation and fuel properties. Under some conditions the presence of soot itself in the combustion field can intrude on this effort. Measurements that rely on digital video imaging of the fuel burning process, such as employed in the FLame Extinguishment eXperiment (FLEX), are vulnerable to this concern. FLEX was developed to understand mechanisms that control flame extinction as it relates to cabin fire safety aboard manned spaceflight, though the broad methodologies are more widely applicable to combustion of fuel systems in combustion engines. The configuration of isolated droplets burning with near spherical symmetry was employed to elucidate the mechanisms involved. The main diagnostic was digital video imaging. From this information, quantitative measurements of the droplet and flame boundaries were made using suitable computer-based data extraction algorithms that revealed the influence of ambient gas conditions and liquid fuel properties on flame extinction. For some of the fuels employed in FLEX, soot formation was evident (i.e., n-heptane droplets burning in air), which obscured part of the droplet and flame boundary that can compromise data accuracy. The success of the FLEX program assumes the existence of efficient and accurate computer-based algorithms that can overcome the obscuration influence of soot on droplets, are able to track droplets if they move, and can account for deviations from sphericity due to ignition disturbances. However, this is not the case. Commercial image analysis packages are limited in their ability to deal with the presence of soot in the imaging field without significant user input. Failure to address this problem impacts not only deriving full benefit from the FLEX experiments already carried out, but also follow-on studies which would utilize digital video imaging to study droplet combustion dynamics of sooting fuels used in commercial transportation systems. In this proposal we address this problem be developing a diameter estimation refinement algorithm that employs concepts that automate droplet and flame diameter measurements from video images in the presence of soot. The algorithm will be used to re-analyze certain FLEX data on the Physical Sciences Informatics (PSI) system -- results from a proof-of-concept program are included in the proposal for illustration -- and to extract quantitative data from new proposed experiments to study the influence of mixture fraction on blends of bio-derived fuels with an aviation fuel: hydroprocessed renewable jet fuel derived from algae with Jet-A (algal feedstocks have potential for larger yields compared to outer sources) for which soot formation is expected. The outcome will be a robust capability to process digital video images that efficiently and precisely extracts measurements from digital images even in the presence of soot aggregates. The results -- data and software -- will be freely available on various outlets as discussed in the proposal. The project includes a team with expertise in developing algorithms for image analysis (Prof. Anthony Reeves of Cornell's Electrical and Computer Engineering Department) and experience in droplet combustion experimentation (Prof. C. Thomas Avedisian of Cornell's Mechanical and Aerospace Engineering (MAE) Department). The project team has supplies of the algae-derived and jet fuel systems that will be studied (Terravia, Inc; and the Air Force).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One of the most important challenges of our time is to mitigate the harmful impacts of particulate emissions (soot) from combustion engines on global climate change, and to reduce the consumption of liquid petroleum fuels that power transportation systems. A promising concept is to mix petroleum fuels with bio-derived fuels. An evaluation of the performance benefits rests on diagnostics to unravel the complex relationships between soot formation and fuel properties. Under some conditions the presence of soot itself in the combustion field can intrude on this effort. Measurements that rely on digital video imaging of the fuel burning process, such as employed in the FLame Extinguishment eXperiment (FLEX), are vulnerable to this concern. FLEX was developed to understand mechanisms that control flame extinction as it relates to cabin fire safety aboard manned spaceflight, though the broad methodologies are more widely applicable to combustion of fuel systems in combustion engines. The configuration of isolated droplets burning with near spherical symmetry was employed to elucidate the mechanisms involved. The main diagnostic was digital video imaging. The primary goal of this research is to develop precision computer algorithms that can measure the diameter of near-spherical fuel droplets in each video image. In previous work general purpose semi-automated measurement tools have been used which suffer to main drawbacks: First the standard algorithms often fail in the presence of soot and second there is no error confidence interval determined for any measurement so the precision of these measurements is unknown. At the outcome we will have a general purpose fully automated image evaluation system that will provide the precision of the diameter measurement for each image. This tool may be applied to any situation where a precise shape measurement of a moving object in a video image. A second goal is to explore mixtures of recent bio-derived fuels that were not considered in the FLEX2 experiments. These are conducted in a drop tower experimental system at Cornell. This work will provide insight in the advantages of the new mixtures. While these are currently analyzed using traditional semi-automated methods we plan to reanalyze the data once the automated precision methods are validated for this data.
Task Progress:	This brief report summarizes research carried out under NASA Grant #80NSSC18K0480. The development of a new image analysis algorithm was accomplished for measuring droplet diameter in the presence of soot formed during the burning process. The algorithm was applied to digital video images from experiments carried out as part of the Flame Extinguishment Experiment-2 (FLEX2) on the International Space Station (ISS). Some of the FLEX2 fuel systems had a strong propensity to produce soot. The presence of soot aggregates significantly impacted the ability to obtain quantitative data from video images of the droplet burning process. We developed an image analysis algorithm in response to this challenge. The FLEX2 fuel systems investigated were 'surrogates' for petroleum-based fuels used in terrestrial propulsion systems (fuels for automotive and jet engines). These were principally binary mixtures of hydrocarbons as noted in the report and all had a propensity to produce soot in various degrees. Soot formation can be a prime concern in the measurement of droplet diameter from video images because many practical hydrocarbon fuels have a high sooting propensity. The algorithm developed as part of the project is able to reconstruct sections of a droplet boundary that are obscured by soot. With data resulting from such an ability that extends current capabilities of image analysis where no data could be obtained, a more complete understanding of

	droplet burning is enabled. [Ed. note 11/3/21: compiled from PI's final progress report sent in July 2021]
Bibliography Type:	Description: (Last Updated: 06/13/2025)
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